

Vulnerability Assessment Report

Cyber Security Task 1 (2026)

By Future Interns

Vulnerability Assessment Report for a Live Website

Prepared By

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Assessment Type

Passive Vulnerability Assessment (Read-Only Scope)

Date

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Table of Contents

1. Executive Summary
 2. Objective
 3. Scope of Assessment
 4. Ethical Guidelines
 5. Target Website
 6. Tools Used
 7. Methodology
 8. Findings Summary
 9. Detailed Findings
 10. Recommendations
 11. Conclusion
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1. Executive Summary

This report presents the results of a passive vulnerability assessment conducted on a publicly accessible website. The purpose of the assessment was to identify common security weaknesses without performing any intrusive or harmful activities.

The assessment focused on:

- Security headers

- Configuration issues
- Public exposure analysis
- Client-side security observations

No exploitation, brute force attacks, denial-of-service attacks, or unauthorized access attempts were performed during this assessment.

The findings identified several security weaknesses that could increase the risk of attacks if left unaddressed. Recommendations have been provided to improve the overall security posture of the website.

2. Objective

The objective of this assessment was to:

- Analyze a public website for common security weaknesses
 - Identify visible security misconfigurations
 - Classify risks based on severity
 - Provide remediation recommendations
 - Prepare a professional vulnerability assessment report
-

3. Scope of Assessment

Included in Scope

- Public-facing web pages
- Passive vulnerability scanning
- Security header analysis
- Configuration review
- Browser-based inspection

Excluded from Scope

- Exploitation attempts
 - Login bypass testing
 - Brute force attacks
 - Denial-of-Service (DoS)
 - Malware deployment
 - Any destructive testing
-

4. Ethical Guidelines

This assessment followed responsible and ethical cybersecurity practices.

The testing was limited to read-only analysis techniques and did not attempt to disrupt, exploit, or damage the target website.

5. Target Website

Website Tested

<https://demo.testfire.net/>

Website Type

Public demo banking application

6. Tools Used

Tool	Purpose
Nmap	Port and exposure analysis
OWASP ZAP	Passive vulnerability scanning
Browser DevTools	Header and cookie inspection
Canva	Report design and formatting

Port & Service Analysis (Nmap)

Description:

A basic Nmap scan was performed to identify exposed services.

Evidence:

```
kali@kali ~$ nmap demo.testfire.net
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-18 10:26 -0400
Nmap scan report for demo.testfire.net (65.61.137.117)
Host is up (0.041s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp    open  https
8080/tcp   open  http-proxy

Nmap done: 1 IP address (1 host up) scanned in 17.88 seconds

kali@kali ~$
```

Observation:

The server exposes standard web ports such as HTTP (80) and HTTPS (443).

7. Methodology

The following methodology was used during the assessment:

- 1. Selected a publicly accessible demo website.
- 2. Performed passive reconnaissance.
- 3. Conducted security header analysis.
- 4. Reviewed publicly visible configurations.
- 5. Captured screenshots of findings.
- 6. Classified risks based on severity.
- 7. Prepared remediation recommendations.

8. Findings Summary

ID	Finding	Risk Level
1	Missing Security Headers	Medium

2	Information Disclosure	Low
3	Missing Content Security Policy	Medium
4	Cookie Security Weaknesses	Medium
5	Outdated Server Information Exposure	Low

9. Detailed Findings

Finding 1: Missing Security Headers

Risk Level

Medium

Description

The application was found to be missing important HTTP security headers that help protect users against browser-based attacks.

Examples include:

- X-Frame-Options
- X-Content-Type-Options
- Strict-Transport-Security

Impact

Missing security headers can increase the risk of:

- Clickjacking attacks
- MIME-type confusion attacks
- Insecure browser communication

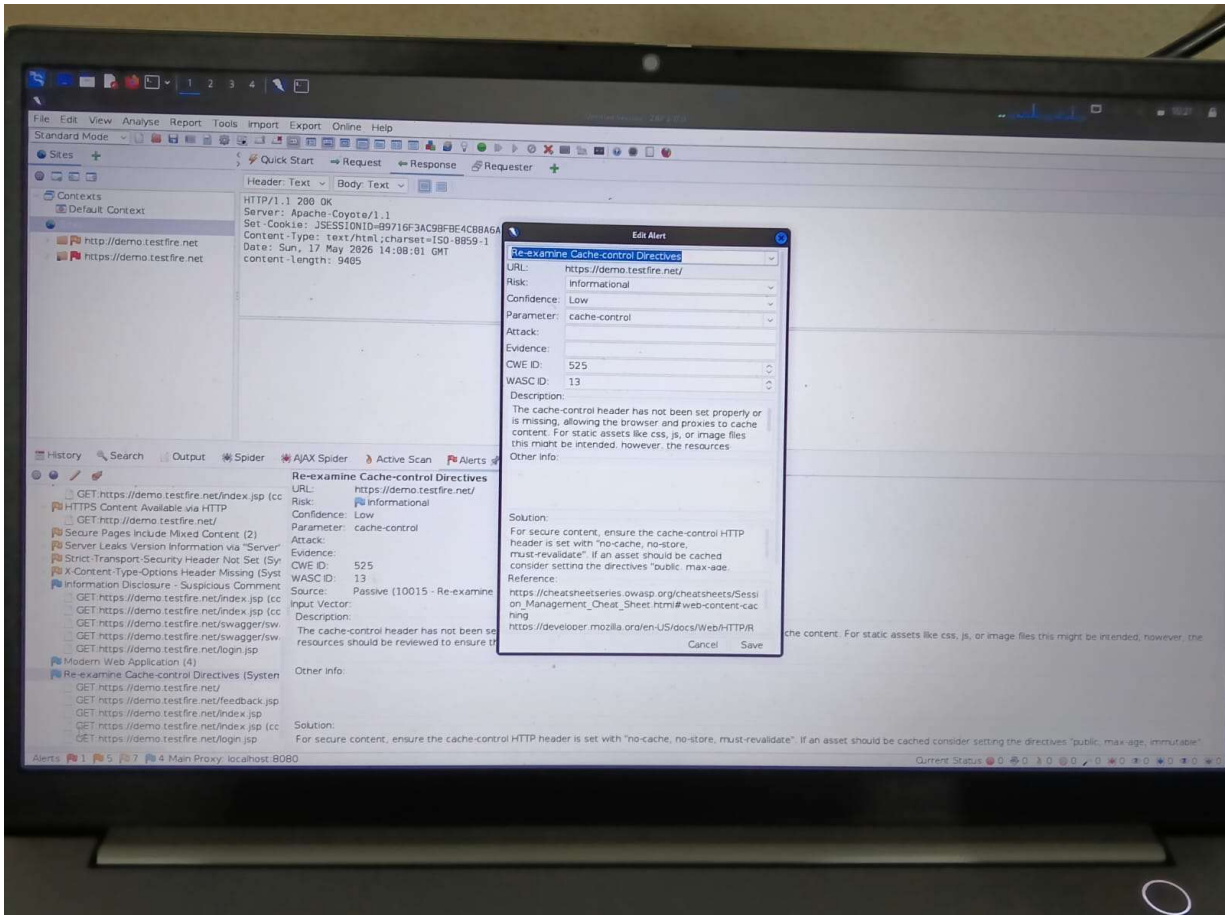
Recommendation

Configure the web server to implement recommended security headers.

Recommended headers include:

```
X-Frame-Options: SAMEORIGIN
X-Content-Type-Options: nosniff
Strict-Transport-Security: max-age=31536000
```

Evidence



Finding 2: Information Disclosure

Risk Level

Low

Description

The application exposed server or framework-related information through HTTP responses.

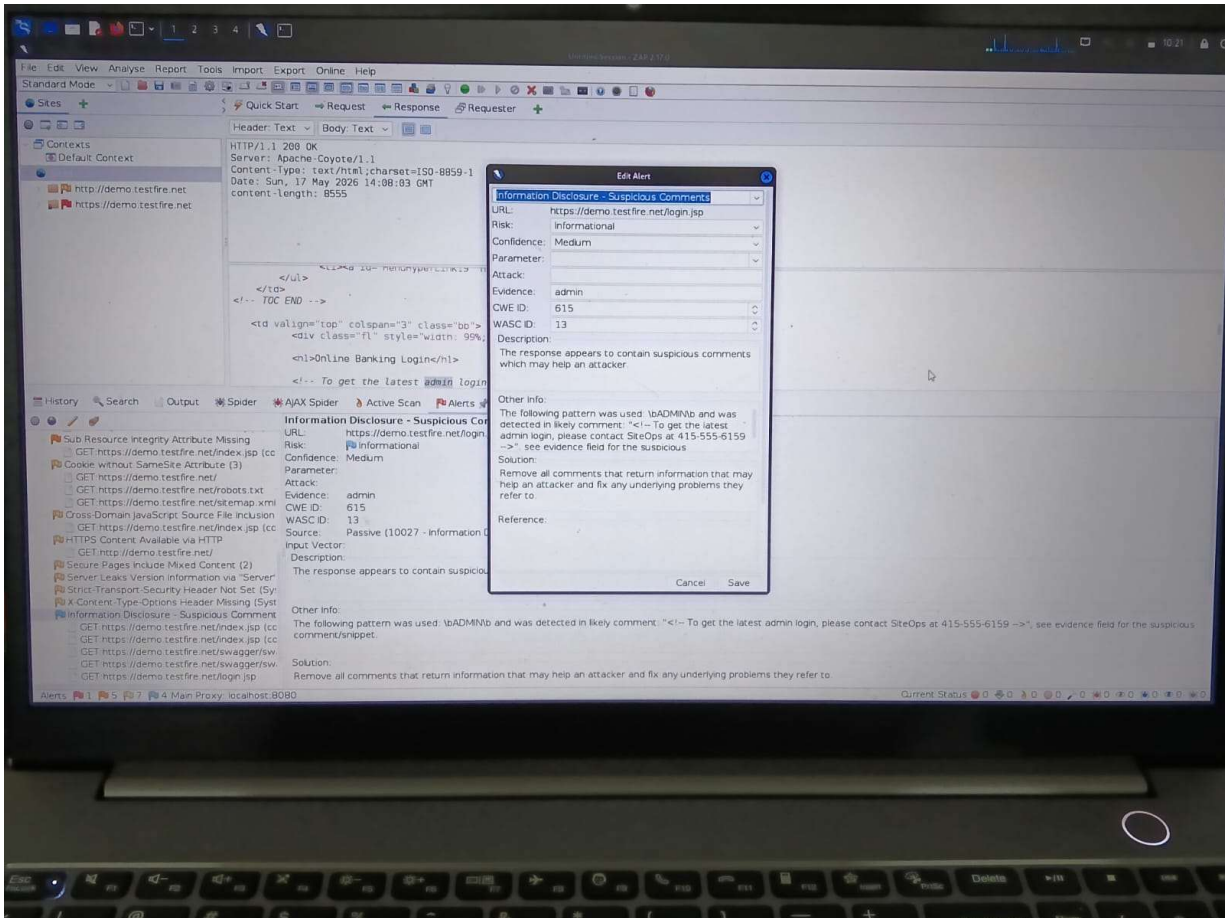
Impact

Attackers may use exposed information to identify technologies and target known vulnerabilities.

Recommendation

- Remove unnecessary server banners
- Hide framework version details
- Disable verbose error messages

Evidence



Impact

Without CSP protection, the website may be more vulnerable to:

- Cross-Site Scripting (XSS)
- Malicious script injection
- Unsafe third-party resource loading

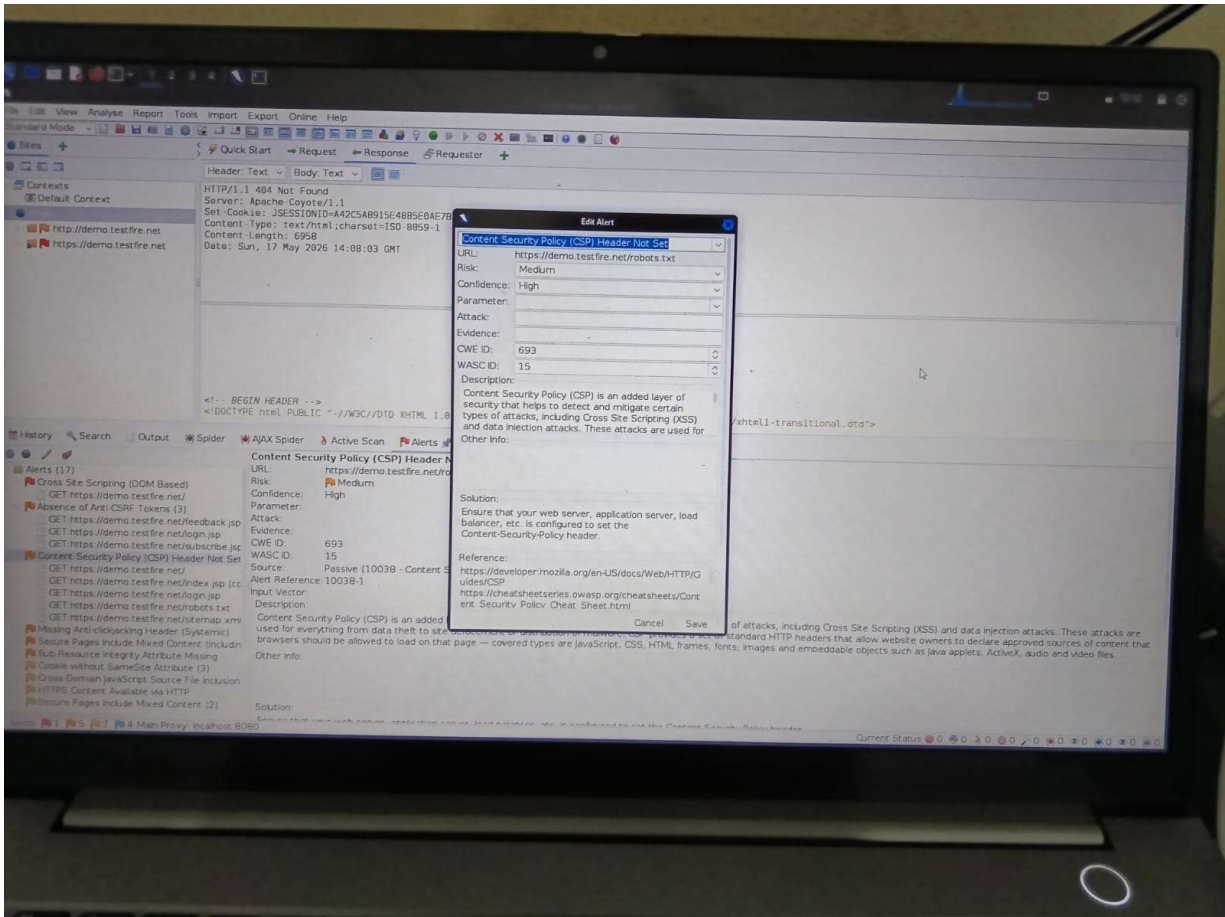
Recommendation

Implement a properly configured Content-Security-Policy header.

Example:

```
Content-Security-Policy: default-src 'self'
```

Evidence



Finding 4: Cookie Security Weaknesses

Risk Level Medium

Description

Some cookies were found without recommended security attributes.

Missing attributes may include:

- Http Only
- Secure
- SameSite

Impact

Improperly configured cookies may increase the risk of:

- Session hijacking
- Client-side script access
- Cross-site request forgery attacks

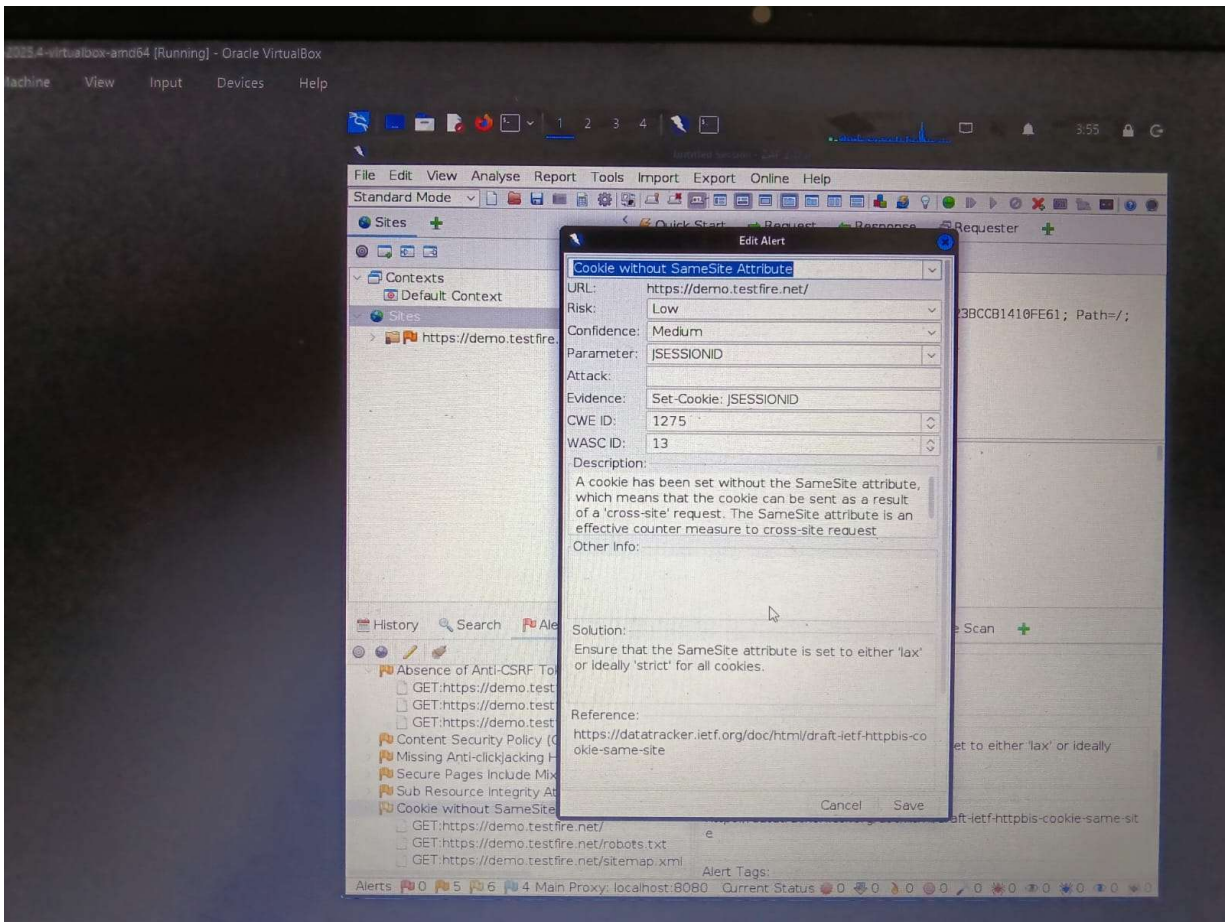
Recommendation

Configure cookies with secure attributes.

Example:

```
Set-Cookie: sessionid=xyz; HttpOnly; Secure; SameSite Strict
```

Evidence



10. Recommendations

The following actions are recommended to improve the security posture of the application:

- Implement all recommended security headers
- Configure secure cookie attributes
- Remove unnecessary server information disclosure
- Apply regular software and framework updates
- Conduct periodic security assessments
- Monitor logs and suspicious activity

11. Conclusion

The passive vulnerability assessment identified several moderate and low-risk security weaknesses within the target website.

Although no critical vulnerabilities were discovered during this read-only assessment, improvements are recommended to strengthen the website's overall security posture and reduce exposure to common web-based attacks.

The findings and recommendations provided in this report can help improve security practices and reduce future risks.

Supporting Evidence

Add all captured screenshots from:

- OWASP ZAP
 - Browser DevTools
 - Nmap outputs at the end of the report or under each related finding.
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Disclaimer

This assessment was conducted strictly within passive and ethical testing boundaries. No exploitation or unauthorized activities were performed during this assessment.